

2016.0 RANGE ROVER (LG), 206-07

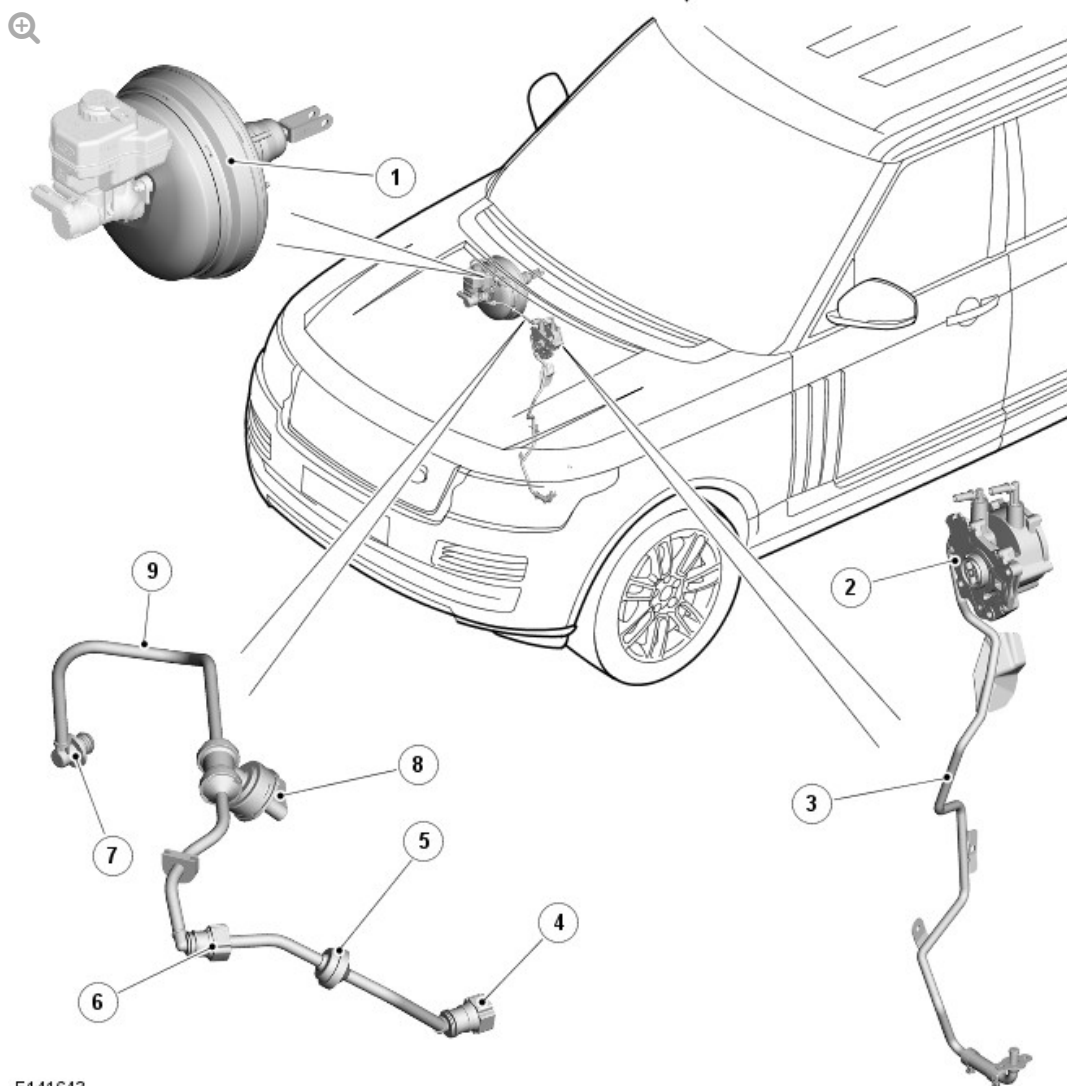
POWER BRAKE ACTUATION

DESCRIPTION AND OPERATION

COMPONENT LOCATION - TDV6 3.0L
DIESEL

NOTE:

RHD (right-hand drive) installation shown, LHD (left-hand drive)
installation similar.



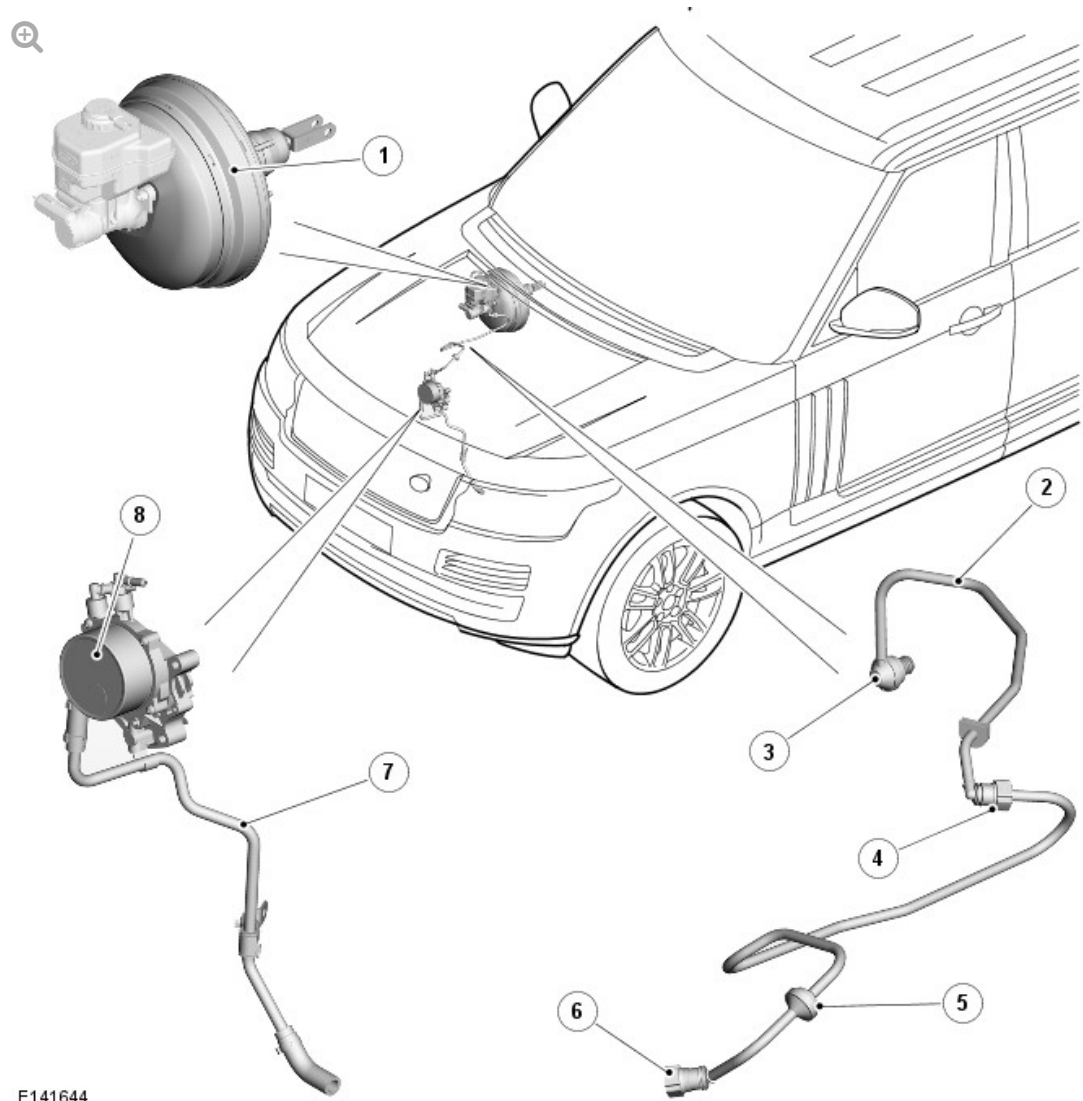
E141643

ITEM	DESCRIPTION
1	Brake booster
2	Vacuum pump
3	Oil scavenge tube
4	Vacuum pump connector
5	Check valve
6	In-line connector
7	Brake booster connector
8	Brake vacuum sensor (with integral check valve)
9	Vacuum pipes

COMPONENT LOCATION - TDV8 4.4L
DIESEL

NOTE:

RHD installation shown, LHD installation similar.



E141644

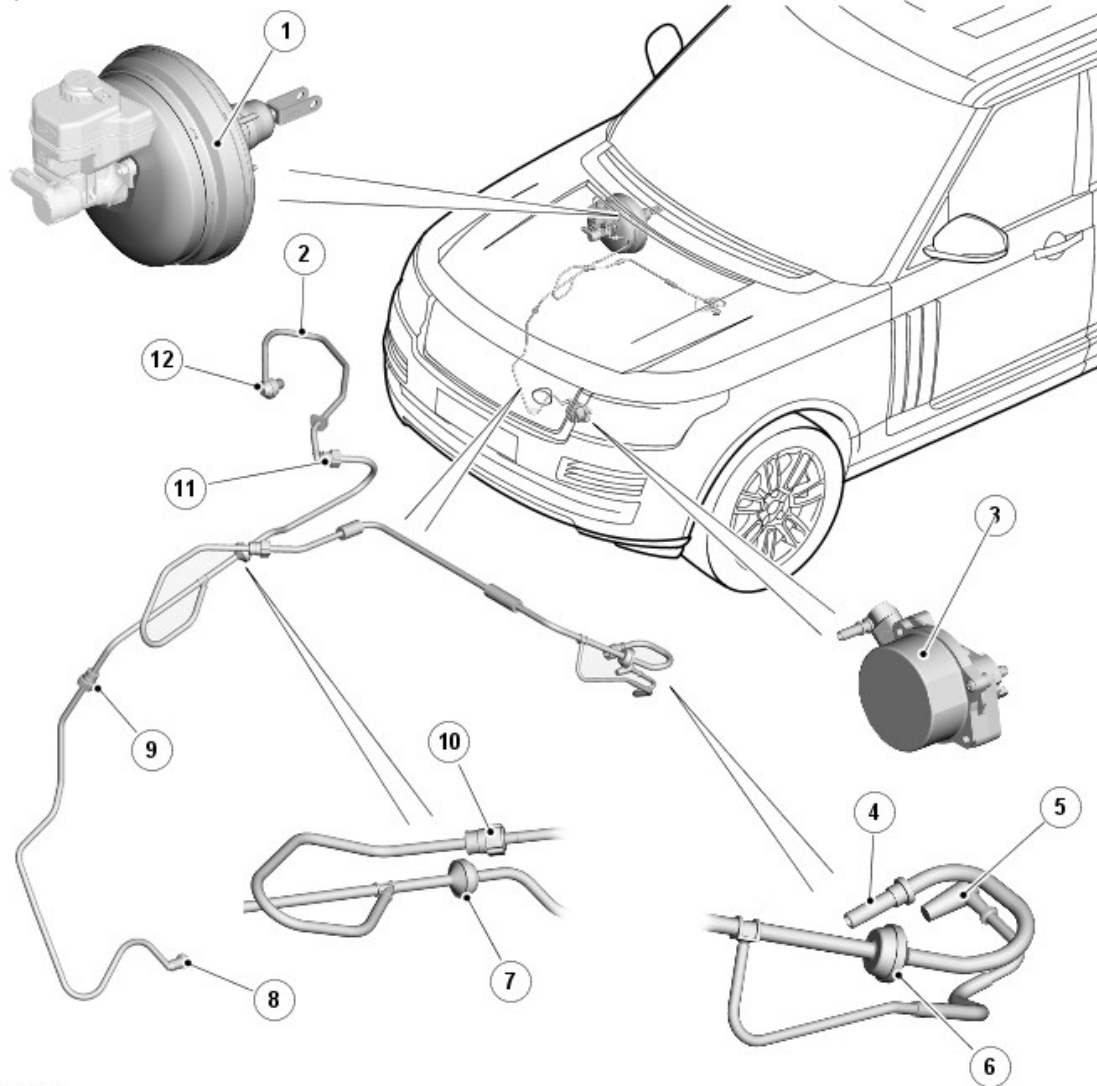
ITEM	DESCRIPTION
1	Brake booster
2	Vacuum pipes
3	Brake booster connector (with integral check valve)
4	In-line connector

5	Check valve
6	Vacuum pump connector
7	Oil scavenge tube
8	Vacuum pump

COMPONENT LOCATION - V8 5.0L NATURALLY ASPIRATED

NOTE:

RHD installation shown, LHD installation similar.



E141645

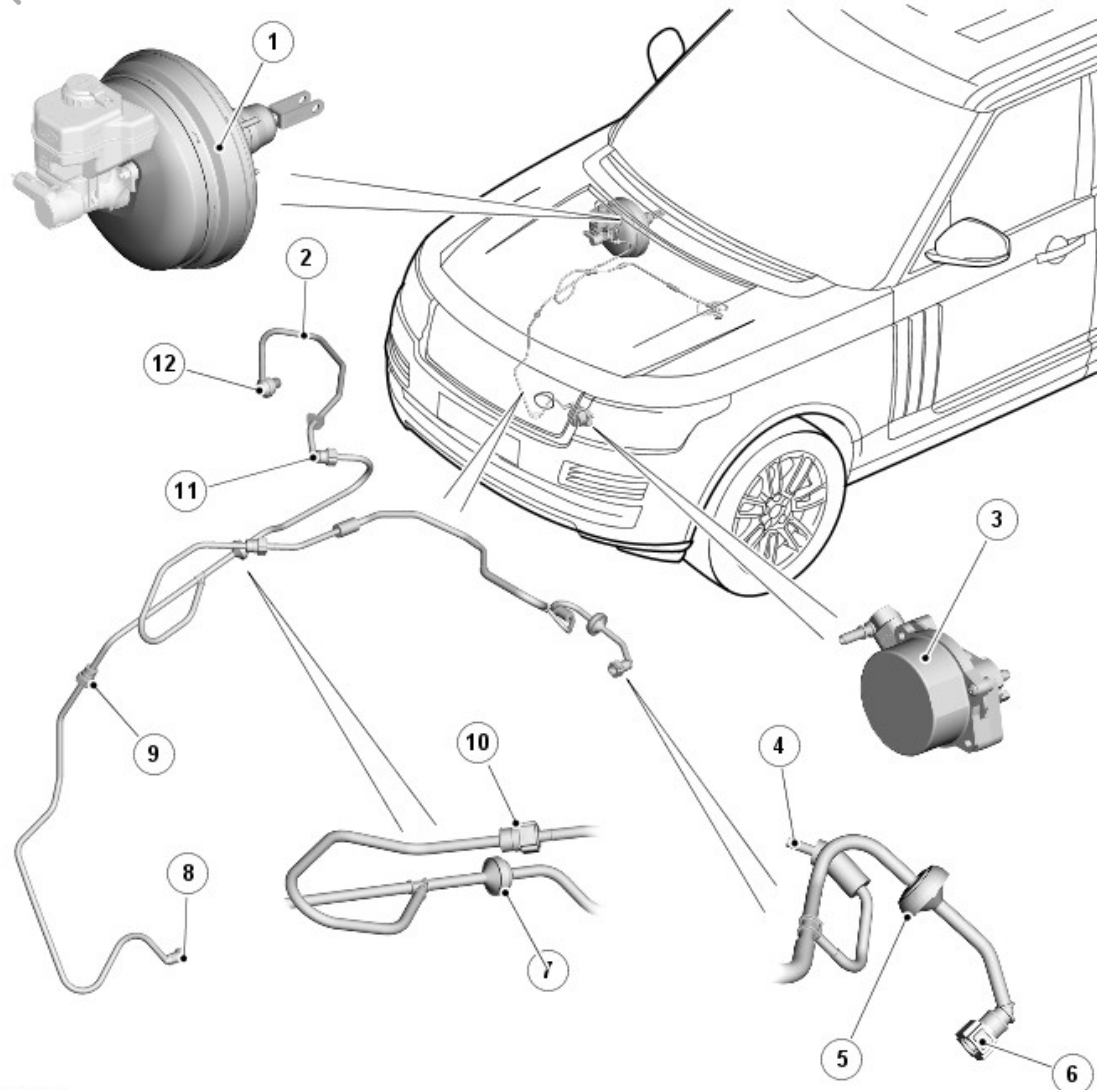
ITEM	DESCRIPTION
1	Brake booster
2	Vacuum pipes
3	Vacuum pump
4	Intake manifold connection
5	Variable intake system connection
6	Check valve
7	Check valve
8	Vacuum pump connector
9	In-line connector

10	In-line connector
11	In-line connector
12	Brake booster connector (with integral check valve)

COMPONENT LOCATION - V8 5.0L SUPERCHARGED AND V6 3.0L SUPERCHARGED

NOTES:

- RHD installation shown, LHD installation similar.
- V8 5.0L S/C shown, V6 3.0L S/C is similar.



E141646

ITEM	DESCRIPTION
1	Brake booster
2	Vacuum pipes
3	Vacuum pump
4	Noise feedback system connection
5	Intake manifold connector
6	Check valve
7	Check valve
8	Vacuum pump connector
9	In-line connector

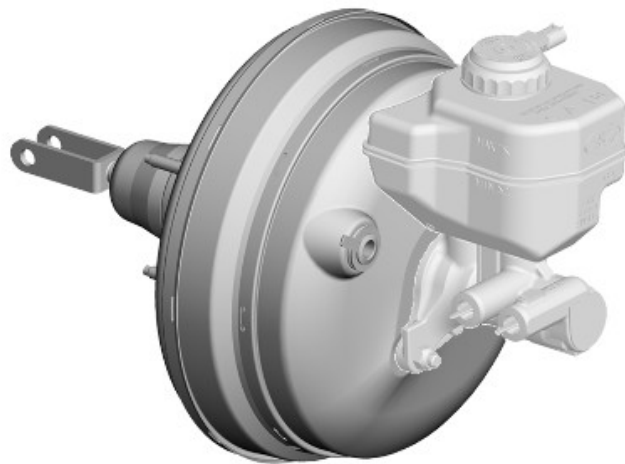
10	In-line connector
11	In-line connector
12	Brake booster connector (with integral check valve)

OVERVIEW

Power assistance for the braking system is provided by a vacuum operated, non-active brake booster. The vacuum required for brake booster operation is provided either by an engine driven vacuum pump (TDV6 3.0L diesel and TDV8 4.4L diesel engines), or a combination of an engine driven vacuum pump and the intake manifold (V8 5.0L and V6 3.0L gasoline engines).

DESCRIPTION

BRAKE BOOSTER



E141647

The brake booster is installed on the driver side of the engine compartment and is sealed to the bulkhead with a gasket. The brake booster actuating push rod is attached to the brake pedal with a clevis pin. The brake booster output rod locates in the primary piston of the brake master cylinder.

The brake booster is a compact unit with tandem diaphragms that operate as a single unit.

VACUUM PIPES

The vacuum pipes consist of rigid plastic piping that connect the vacuum pump and, on V8 5.0L and V6 3.0L gasoline engines, the intake manifold, to the brake booster.

Depending on engine type, one or more check valves are installed in the vacuum pipes to maintain the vacuum in the brake booster. On V8 5.0L and V6 3.0L gasoline engines, a check valve close to the intake manifold prevents air and fuel vapor being drawn into the vacuum pipes when there is minimal depression in the intake manifold.

On V8 5.0L and V6 3.0L gasoline engines, the vacuum pipes incorporate a connection for the variable intake system of the intake manifold.

For additional information, refer to: [Intake Air Distribution and Filtering](#) (303-12C Intake Air Distribution and Filtering - V8 N/A 5.0L Petrol, Description and Operation) /

[Intake Air Distribution and Filtering](#) (303-12D Intake Air Distribution and Filtering - V8 S/C 5.0L Petrol, Description and Operation) /

[Intake Air Distribution and Filtering](#) (303-12B Intake Air Distribution and Filtering - V6 S/C 3.0L Petrol, Description and Operation).

On V8 5.0L and V6 3.0L gasoline engines, the vacuum pipes incorporate a connection for the noise feedback system.

For additional information, refer to: [Intake Air Distribution and Filtering](#) (303-12C Intake Air Distribution and Filtering - V8 N/A 5.0L Petrol, Description and Operation) /

[Intake Air Distribution and Filtering](#) (303-12D Intake Air Distribution and Filtering - V8 S/C 5.0L Petrol, Description and Operation) /

[Intake Air Distribution and Filtering](#) (303-12B Intake Air Distribution and Filtering - V6 S/C 3.0L Petrol, Description and Operation).

BRAKE VACUUM SENSOR - VEHICLES WITH STOP/START ONLY



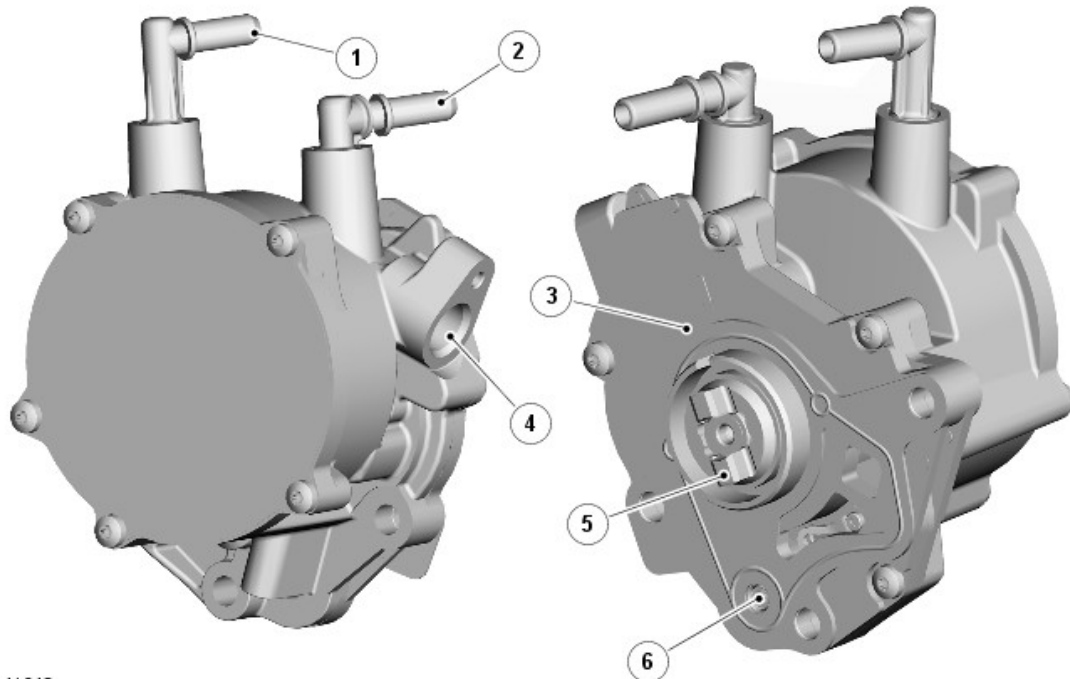
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On vehicles with the stop/start system, the vacuum pipes incorporate a brake vacuum sensor with an integral check valve. The brake vacuum sensor monitors vacuum reserves in the brake booster and is hardwired to the ECM (engine control module).

For additional information, refer to: [Starting System](#) (303-06A Starting System - TDV6 3.0L Diesel, Description and Operation).

VACUUM PUMP

TDV6 3.0L Diesel Engines



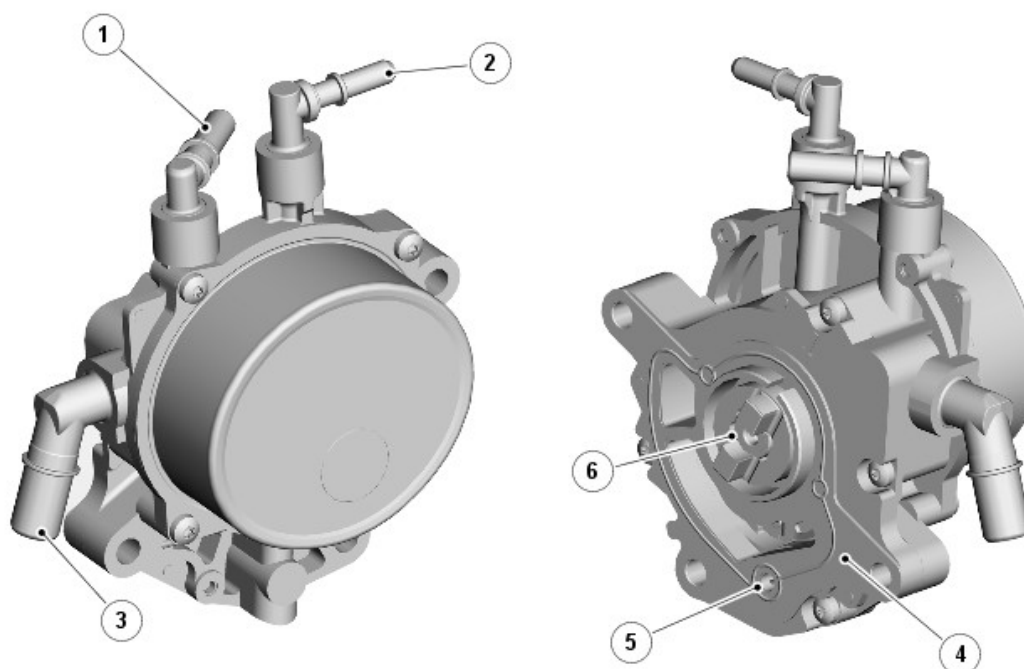
E141648

ITEM

DESCRIPTION

1	Vacuum connection for exhaust gas recirculation/turbocharger control systems
2	Vacuum connection for brake booster
3	Gasket
4	Oil scavenge connection
5	Drive coupling
6	Oil inlet

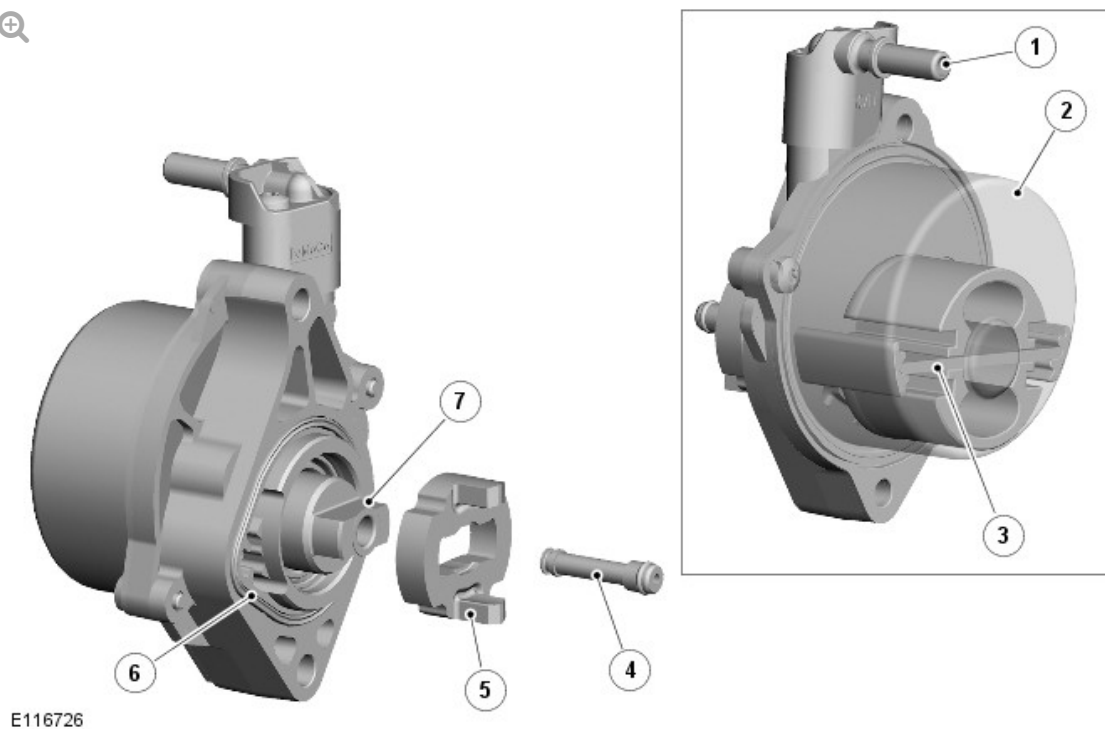
TDV8 4.4L Diesel Engines



E141649

ITEM	DESCRIPTION
1	Vacuum connection for brake booster
2	Vacuum connection for exhaust gas recirculation/turbocharger control systems
3	Oil scavenge connection
4	Gasket
5	Oil inlet
6	Drive coupling

V8 5.0L Naturally Aspirated and Supercharged and V6 3.0L Supercharged Engines



ITEM	DESCRIPTION
1	Brake vacuum pipe connection
2	Cover
3	Vane
4	Oil tube
5	Drive coupling
6	Seal
7	Rotor

On V8 5.0L and V6 3.0L gasoline engines, the vacuum pump is installed on the front of the engine sump body and driven by the auxiliary camshaft. A drive coupling on the pump rotor engages with slots in the front of the auxiliary camshaft. An oil tube in the center of the rotor supplies engine oil from the auxiliary camshaft to lubricate and cool the vacuum pump. Drain oil and air from the brake booster and vacuum pipes is expelled into the

engine sump through the sump body.

On diesel engines, the vacuum pump is installed on the front (TDV8 4.4L engines) or rear (TDV6 3.0L engines) of the right cylinder head and driven by the inlet camshaft. A drive coupling on the pump rotor engages with slots in the end of the inlet camshaft. The vacuum pump is lubricated and cooled by engine oil supplied from a gallery in the cylinder head to an inlet on the mating face of the vacuum pump. Drain oil and air from the brake booster and vacuum pipes is expelled into the camshaft cover, together with oil from the scavenge tube connected to the oil pan.

On diesel engines, the vacuum pump has an additional vacuum connection, which is connected to the EGR (exhaust gas recirculation) system and the turbocharger control system. For additional information, refer to:

[Turbocharger](#) (303-04B Fuel Charging and Controls - Turbocharger - TDV6 3.0L Diesel, Description and Operation),

[Turbocharger](#) (303-04G Fuel Charging and Controls - Turbocharger - TDV8 4.4L Diesel, Description and Operation),

Engine Emission Control (303-08A, Description and Operation),

Engine Emission Control (303-08E, Description and Operation).